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Estimating pollution spread in water networks as a Schrödinger bridge problem with partial information[☆]

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ABSTRACT

Incidents where water networks are contaminated with microorganisms or pollutants can result in a large number of infected or ill persons, and it is therefore important to quickly detect, localize and estimate the spread and source of the contamination. In many of today's water networks only limited measurements are available, but with the current internet of things trend the number of sensors is increasing and there is a need for methods that can utilize this information. Motivated by this fact, we address the problem of estimating the spread of pollution in a water network given measurements from a set of sensors. We model the water flow as a Markov chain, representing the system as a set of states where each state represents the amount of water in a specific part of the network, e.g., a pipe or a part of a pipe. Then we seek the most likely flow of the pollution given the expected water flow and the sensors observations. This is a large-scale optimization problem that can be formulated as a Schrödinger bridge problem with partial information, and we address this by exploiting the connection with the entropy regularized multimarginal optimal transport problem. The software EPANET is used to simulate the spread of pollution in the water network and will be used for testing the performance of the methodology.

1. Introduction

The World Health Organization recognizes access to safe drinking-water as essential to health, a basic human right and a component of effective policy for health protection [23]. At the same time, failure to ensure drinking-water safety may expose the community to the risk of outbreaks of infectious diseases. The possible sources of contamination include pathogens and toxic chemicals, which can potentially enter water distribution systems as a result of malicious attacks, infrastructures aging or natural disasters [6,7,15,29]. The traditional laboratory-based methods for detection of pollution in water networks do not provide real-time public health protection. A significant measure that can be applied in order to mitigate the consequences of a contamination is the strategic placement of online water quality sensors [2]. The parameters that are typically measured for early warning systems include, among others, pH, chlorine, temperature, flow and turbidity [33]. It has been shown that the variation of such parameters can be used to detect various types of chemical and biological contaminants [13,21]. Furthermore, recently, a large number of low cost sensors are becoming available [18,27]. The availability of real-time sensor measurements allows for the development of applications in a large number of areas, ranging from detection of leakage and sensor errors [20,32], and real-time control of reservoirs [16] to applications in cyberphysical security [24]. In addition to these newly developed methods there are also public domain software packages, such as EPANET (Environmental Protection Agency Network Evaluation Tool) [28], that can be used for real-time simulation of water networks.

Online sensing and real-time computations open up for development of new solutions for detection and estimation of pollution in water distribution systems, and with the aim of using the data produced by these sensors we propose a novel approach to estimate the spread of pollution in water networks, given the measurements from a set of sensors. Our approach is based on Markov modelling of the water diffusion in networks, where the states of the Markov chain are pipe segments. A related approach has

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been developed to model the spread of pathogen particles in the air [5]. In that case, an indoor environment has been divided into boxes, which correspond to the states of the Markov chain. Once we model the water network as a Markov chain, we aim to find the most likely flow of pollution given the expected water flow and the sensors observations. This results in a large-scale optimization problem, which we formulate as a Schrödinger bridge problem with partial information. The Schrödinger bridge problem is a classical problem in statistical mechanics [19,30]. Given two observations of a particle distribution at two time instances, and a prior on the particle evolutions (in the classical setting this is the Brownian motion), the Schrödinger bridge describes the most likely evolution of the particle distributions. It turns out that the most likely evolution can be characterized as the one that minimizes the relative entropy to the prior, while also satisfying the observations. A discretized version of the Schrödinger bridge problem is based on modelling the particle evolutions as a Markov chain [11,25]. The problem can be efficiently solved by an iterative scheme, which is linked to the so called Sinkhorn iterations for the optimal transport problem [1,9,12]. Optimal transport methods have previously been used for inference problems [31], and they have shown to provide a robust metric in hydrologic systems [22].

In this work we utilize the discrete Schrödinger bridge to model the spread of pollution particles in a water network. We augment the framework to the settings, where only partial observations of the particle distributions are known. This corresponds to the fact that observations of the pollution spread can only be made at a few locations in the network, where sensors are placed. We solve the obtained optimization problem by developing a computational method for the Schrödinger bridge problem with partial information.

The outline of the paper is as follows: Section 2 provides background material on the Schrödinger bridge problem. In Section 3, which contains the main contribution, we formulate the Schrödinger bridge problem in the case of partial observations, and we develop an iterative algorithm for solving it. In Section 4 we explain how water networks can be modelled as Markov chains. Then, in Section 5 we combine the results of the previous two sections to provide an example of application of the Schrödinger bridge problem to a water network with sensors observations.

2. Background

2.1. Notation

By $\odot, \oslash, \exp(\cdot), \log(\cdot)$ we denote element-wise multiplication, division, exponential and logarithm of matrices and vectors. With $\mathbb{1}$ we denote a column vector of ones of adequate size, $\mathrm{supp}(\cdot)$ indicates the support of a matrix, i.e., the non-zero elements.

2.2. Schrödinger bridge problem

In this section we introduce the discrete Schrödinger bridge problem, in which particle dynamics are modelled as a Markov chain [11,25]. More precisely, we assume that, given a cloud of indistinguishable particles, each of them evolves over a set of n states $X = \{X_1, X_2, \dots, X_n\}$. Let q_t denote the state of a particle at time t. Then the underlying probability law is given by a state transition matrix $A = \left[a_{ij}\right]_{i,j=1}^n$, where $a_{ij} = P(q_{t+1} = X_j | q_t = X_i)$.

Consider two observed particle distributions, described by vectors $\mu_t \in \mathbb{N}^n$ for time $t \in \{0, 1\}$. Here the ith element $(\mu_t)_i$ denotes the number of particles in state X_i at time t. The particle evolution is characterized by a matrix $M = [m_{ij}]_{i,j=1}^n$, where m_{ij} denotes the number of particles that transit from state X_i to state X_j.

Given the initial distribution μ_0 and the transition probability matrix A, the probability that particles transition according to the evolution matrix M is

$$P_{\mu_0,A}(M) = \prod_{i=1}^n \left(\binom{(\mu_0)_i}{m_{i1}, m_{i2}, \dots, m_{in}} \prod_{j=1}^n a_{ij}^{m_{ij}} \right), \tag{1}$$

where $\binom{\cdot}{\cdot,\dots,\cdot}$ denotes a multinomial coefficient. It can be shown that if the number of particles is large, then the log-likelihood of (1) can be approximated in terms of a KL divergence.

Definition 1. Let p and q be two nonnegative vectors or matrices of the same dimension. The normalized Kullback–Leibler (KL) divergence between p from q is defined as

$$H(p|q) := \sum_i \left(p_i \log\left(\frac{p_i}{q_i}\right) - p_i + q_i \right),$$

where $0 \log 0$ is defined to be 0.

Proposition 1 ([11]). *Given the transition matrix A, let $\mu_0^{(N)} \in \mathbb{N}^n$ be a sequence of distributions with N particles, and $M^{(N)} \in \mathbb{N}^{n\times n}$ be a sequence of mass transfer matrices such that $M^{(N)}\mathbb{1} = \mu_0^{(N)}$ and $\mathrm{supp}(M^{(N)}) \subseteq \mathrm{supp}(\mathrm{diag}(\mu_0^{(N)})A)$. Then there exists a constant $C > 0$ such that for all N it holds that*

$$\left| \log\left(P_{\mu_0^{(N)},A}(M^{(N)}) \right) + H\left(M^{(N)} \,\middle|\, \mathrm{diag}(\mu_0^{(N)})A\right) \right| \le C \log(N).$$

□

This means that as the number of particles becomes large, the negative log-likelihood of a given transition can be approximated by the relative entropy,

$$P_{\mu_0^{(N)},A}(M^{(N)}) \approx e^{-H\left(M^{(N)} \,\middle|\, \mathrm{diag}(\mu_0^{(N)})A\right)}.$$

Note that the relative entropy expression allows us to consider particles as continuous quantities, i.e., $M \in \mathbb{R}_+^{n\times n}$. This represents the fact that, as the number of particles becomes large, the discrete particle distributions can be approximated by densities. A feasible evolution matrix between the two particle distributions $\mu_0 \in \mathbb{R}_+^n$ and $\mu_1 \in \mathbb{R}_+^n$ must satisfy $M\mathbb{1} = \mu_0$ and $M^T\mathbb{1} = \mu_1$. Thus, the most likely evolution matrix M between the two distributions can be approximated by the solution to

$$\begin{aligned} \underset{M\in\mathbb{R}^{n\times n}}{\text{minimize}} \quad & H(M \,|\, \mathrm{diag}(\mu_0)A) \\ \text{subject to} \quad & M\mathbb{1} = \mu_0, \quad M^T\mathbb{1} = \mu_1. \end{aligned} \tag{2}$$

The Schrödinger bridge problem can be extended to a Markov chain of length τ. Then, given the distributions μ_0 and μ_τ, at time 0 and τ, respectively, we can find the most likely evolution of the particles between them by solving

$$\begin{aligned} \underset{\substack{M^{[0:\tau-1]},\\ \mu_{[1:\tau-1]}}}{\text{minimize}} \quad & \textstyle\sum_{t=1}^{\tau} H(M^t \,|\, \mathrm{diag}(\mu_t)A) \\ \text{subject to} \quad & M^t\mathbb{1} = \mu_t, \quad (M^t)^T\mathbb{1} = \mu_{t+1}, \\ & \text{for } \ t = 0, \dots, \tau-1. \end{aligned} \tag{3}$$

This problem can be understood as a time and space discrete form of the Schrödinger bridge [11,12,25].

3. Problem formulation and main results

In this section we will formulate the problem of estimating the pollution spread in a water network using an optimization problem similar to (3) and then, by considering the dual problem, we determine an efficient method for solving it that works also for large scale problems. To this end, consider a water network divided into n states, each representing a pipe-segment. We will model the evolution of the pollution over τ time instances, and we assume that each pollution particle evolves independently of each other and according to a Markov chain over these n states. Then we may use the same model as in Section 2.2. We indicate with A^t the state transition matrix at time t, which is assumed to be know, and may be estimated from the water-flow. Moreover, let $M^t = [m^t]_{ij}$ describe the flow of pollution, i.e., m^t_{ij} indicates the amount of pollution that transitions from state i to state j between the times t and $t+1$. We assume that sensors at a few given locations (states) in the water network measure the amount of pollution, and these states are denoted by $\pi = (\pi_1, \pi_2, \dots, \pi_k)$. The measured pollution levels at these k states are described by vectors $\rho_t \in \mathbb{R}^k$, for $t \geq 1$. We introduce the matrix $B = [e_{\pi_1}, \dots, e_{\pi_k}]^T \in \{0,1\}^{k\times n}$, where e_i is the ith (column) vector of the canonical basis. Then, the matrix B has a nonzero column i if state i is known for all times. Here, also let $\mathbb{1}_C \in \{0,1\}^n$ denote the vector with ones in the unknown states, i.e., $\mathbb{1}_C = (I - B^T B)\mathbb{1}$. In addition, we assume that we have complete information of the pollution at the initial time, denoted by $\rho_0 \in \mathbb{R}^n$. Now we formulate the following problem with a similar structure as (3):

$$\underset{M^{[0:\tau-1]}}{\text{minimize}} \quad \sum_{t=0}^{\tau-1} H(M^t \mid \operatorname{diag}(M^t\mathbb{1})A^t) \tag{4a}$$

$$\text{subject to} \quad M^0\mathbb{1} = \rho_0, \tag{4b}$$

$$BM^t\mathbb{1} = \rho_t \qquad \text{for} \quad t = 1, \dots, \tau-1, \tag{4c}$$

$$B(M^{\tau-1})^T\mathbb{1} = \rho_\tau, \tag{4d}$$

$$M^t\mathbb{1} = (M^{t-1})^T\mathbb{1} \qquad \text{for} \quad t = 1, \dots, \tau-1. \tag{4e}$$

In particular, this optimization problem extends problem (3) by adding partial constraints on the distributions at time instances $t > 1$. Thus, it can be seen as a Schrödinger bridge problem, where at each time instance partial information of the particle distributions is available. As an extension to the Schrödinger bridge framework, we believe that problem (4) may be of general interest to the control community. In particular, we envision applications to various other settings of estimation and control for large group of agents, where partial state information is given, cf. [10,31]. An advantage of the Schrödinger bridge framework is that it requires only a few model assumptions, and its optimal solution is robust to errors in the assumed underlying model defined by the transition probabilities A^t [11,26].

Next we derive an algorithm for solving (4), which is based on a dual block coordinate ascent method.

Theorem 1. *A dual of the minimization problem (4) is given by*

$$\underset{\lambda^{[0:\tau]},\nu^{[1:\tau-1]}}{\text{maximize}} \quad \sum_{t=0}^{\tau} \lambda_t^T\rho_t - (\rho_0 \odot \exp(\lambda_0))^T A^0 \exp(\nu_1) \tag{5a}$$

$$\text{subject to} \quad \exp(B^T\lambda_{\tau-1}) \odot A^{\tau-1}\exp(B^T\lambda_\tau) \leq \exp(\nu_{\tau-1}), \tag{5b}$$

$$\exp(B^T\lambda_t) \odot A^t\exp(\nu_{t+1}) \leq \exp(\nu_t) \quad \text{for} \quad t = 1, \dots, \tau-2, \tag{5c}$$

where λ_0, λ_t, λ_τ, ν_t are the Lagrangian multipliers corresponding respectively to constraints (4b), (4c), (4d), and (4e).

Proof. First note that the objective function (4a), due to the constraint (4b), can be rewritten as

$$H(M^0 \mid \operatorname{diag}(\rho_0)A^0) + \sum_{t=1}^{\tau-1} H(M^t \mid \operatorname{diag}(M^t\mathbb{1})A^t). \tag{6}$$

Then we relax all the constraints, indicating with λ_0, λ_t $(t = 1, \dots, \tau-1)$, λ_τ, ν_t $(t = 1, \dots, \tau-1)$ the Lagrangian multipliers corresponding to the constraints (4b), (4c), (4d), and (4e), respectively. We obtain the Lagrangian

$$\begin{aligned} L(M,\lambda,\nu) = {} & H(M^0 \mid \operatorname{diag}(\rho_0)A^0) \\ & + \sum_{t=1}^{\tau-1} H(M^t \mid \operatorname{diag}(M^t\mathbb{1})A^t) + \lambda_0^T(\rho_0 - M^t\mathbb{1}) \\ & + \sum_{t=1}^{\tau-1} \lambda_t^T(\rho_t - BM^t\mathbb{1}) + \lambda_\tau^T(\rho_\tau - B(M^{\tau-1})^T\mathbb{1}) \\ & + \sum_{t=1}^{\tau-1} \nu_t^T(M^t\mathbb{1} - (M^{t-1})^T\mathbb{1}). \end{aligned} \tag{7}$$

Taking the derivative with respect to M^t_{ij} and setting it equal to zero, we get for $t = 1, \dots, \tau-2$ that

$$\frac{\partial L}{\partial M^t_{ij}} = \log\left(\frac{M^t_{ij}}{A^t_{ij}(M^t\mathbb{1})_i}\right) - (B^T\lambda_t)_i + (\nu_t)_i - (\nu_{t+1})_j = 0. \tag{8}$$

For the derivative of the entropy term we have used that A^t is row stochastic for $t = 0, \dots, \tau-1$. This leads to

$$M^t_{ij} = (M^t\mathbb{1})_i A^t_{ij} \exp((B^T\lambda_t)_i - (\nu_t)_i + (\nu_{t+1})_j), \tag{9}$$

for $t = 1, \dots, \tau-2$, which in matrix form also can be written as,

$$M^t = \operatorname{diag}(M^t\mathbb{1} \odot \exp(B^T\lambda_t) \oslash \exp(\nu_t))A^t \operatorname{diag}(\exp(\nu_{t+1})). \tag{10}$$

Proceeding in the same way we obtain, for $t = 0$,

$$M^0 = \operatorname{diag}(\rho_0 \odot \exp(\lambda_0))A^0 \operatorname{diag}(\exp(\nu_1)), \tag{11}$$

while, for $t = \tau-1$, we get

$$\begin{aligned} M^{\tau-1} = {} & \operatorname{diag}\left((M^{\tau-1}\mathbb{1}) \odot \exp(B^T\lambda_{\tau-1}) \oslash \exp(\nu_{\tau-1})\right) \\ & \times A^{\tau-1} \operatorname{diag}\left(\exp(B^T\lambda_\tau)\right). \end{aligned} \tag{12}$$

Since the problem is convex, an optimal set of variables $\{M^t\}_t$ satisfies (10)–(12) if they exist. Existence of such an optimal solution is ensured by

$$\begin{aligned} & \exp(B^T\lambda_{\tau-1}) \oslash \exp(\nu_{\tau-1})) \odot A^{\tau-1}\exp(B^T\lambda_\tau) \leq \mathbb{1} \\ & \exp(B^T\lambda_t) \oslash \exp(\nu_t)) \odot A^t\exp(\nu_{t+1}) \leq \mathbb{1} \end{aligned} \tag{13}$$

for $t = 1, \dots, \tau-2$, and one can see that this must hold by multiplying (10) or (12) by $\mathbb{1}$. Note that strict inequality can only hold when the corresponding entries of M^t are zero. If (13) does not hold, one can find a (non-negative) descent direction for one of the variables M^t for which the objective converges to $-\infty$.

Finally, plugging (10)–(12) into the Lagrangian (7), we obtain that

$$L(M^*,\lambda,\nu) = \sum_{t=0}^{\tau} \lambda_t^T\rho_t - (\rho_0 \odot \exp(\lambda_0))^T A^0 \exp(\nu_1), \tag{14}$$

where we ignored the constant term $\rho_0^T\mathbb{1}$ that result from the normalizing terms in the definition of entropy. By maximizing (14) with respect to λ_t and ν_t, together with constraints (13), we obtain the dual problem (5). □

Note that, since ρ_0 is known, the optimal transport matrices M^t, for $t = 0, 1, \dots, \tau$, can be sequentially computed from (11), (10), and (12), for any set of dual variables, making use of the constraints (4e). Moreover, the dual formulation can be simplified by eliminating the variables ν_t for $t = 1, \dots, \tau$, as described by the following Corollary.

Corollary 1. *The dual problem (5) can be equivalently formulated as:*

$$\max_{\lambda_t^{[0:\tau]}} \sum_{t=0}^{\tau} \lambda_t^T\rho_t - \rho_0^T \operatorname{diag}(u_0)A^0 \operatorname{diag}(u_1)\dots\operatorname{diag}(u_{\tau-1})A^{\tau-1}u_\tau, \tag{15}$$

where $u_0 = \exp(\lambda_0)$ *and* $u_t = \exp(B^T\lambda_t)$ *for* $t = 1, \dots, \tau$.

Proof. We will show that the problems are equivalent in the sense that for any fixed set of variables λ_t $(t = 0, \dots, \tau)$, the objective value (15) corresponds to the objective of (5) with the feasible variables ν_t $(t = 1, \dots, \tau-1)$ that maximize the objective. Note that ν_t only appears for $t = 1$ in the objective function of (5), and since ρ_0 and A^0 are nonnegative, the maximizing choice of ν_t for $t = 2, \dots, \tau-1$ is the one that minimizes ν_1 elementwise. This is achieved by selecting ν_t in such a way that equality holds first for $t = \tau-1$ in (5b) and then for $t = \tau-2, \tau-3, \dots, 1$ in (5c). With this selection, we recursively define the variables $w_t := \exp(\nu_t)$ for $t = 1, \dots, \tau-1$ as

$$w_{\tau-1} = u_{\tau-1} \odot A^{\tau-1} u_\tau, \tag{16}$$

and

$$w_t = u_t \odot A^t w_{t+1}, \quad t = 1, \dots, \tau-2, \tag{17}$$

where $u_0 := \exp(\lambda_0)$ and $u_t := \exp(B^T\lambda_t)$, for $t = 1, \dots, \tau$. Using these relations we can replace the term $\exp(\nu_1) = w_1$ and write the objective (5a) only as a function of λ_t, $t = 0, \dots, \tau$. □

The problems (5) and (15) are convex since they are dual problems. However, the convexity can also be seen directly by noting that any term on the form $\exp(\sum_t \eta_t^T \lambda_t)$ is convex. Problem (15) has similarities to the dual of the entropy regularized optimal transport problem (see, e.g., [12]); the latter can be efficiently solved using a block coordinate ascent method. In fact, it turns out that the same approach yields a computationally efficient algorithm that allows for solving problem (15) even in large scale settings.

Proposition 2. *Let* $u_t, \phi_t, \widehat{\phi}_t \in \mathbb{R}^n$ *for* $t = 0, \dots, \tau$ *be positive initial values. Then the solution of the dual problem (15) can be found as the limit point of Algorithm 1.* □

Algorithm 1 Block coordinate ascent for problem (15).

$\hat{\phi}_0 \leftarrow \rho_0$, $\phi_\tau \leftarrow \mathbb{1}$
for $t = \tau-1 : -1 : 0$ **do**
  $\phi_t \leftarrow A^t(u_{t+1} \odot \phi_{t+1})$
end for
while constraint violation (4b)-(4e) $>$ tolerance **do**
  update $u_0 := \rho_0 \oslash (\phi_0 \odot \hat{\phi}_0)$
  for $t = 1 : \tau$ **do**
    $\hat{\phi}_t \leftarrow (A^{t-1})^T(\hat{\phi}_{t-1} \odot u_{t-1})$
    update $\hat{u}_t := \rho_t \oslash B(\phi_t \odot \hat{\phi}_t)$
    $u_t \leftarrow B^T\hat{u}_t + \mathbb{1}_C$
  end for
  for $t = \tau-1 : -1 : 0$ **do**
    $\phi_t \leftarrow A^t(u_{t+1} \odot \phi_{t+1})$
  end for
end while

Proof. We solve (15) by block coordinate ascent. The unconstrained objective function is maximized iteratively with respect to each of the variables $\lambda_0, \dots, \lambda_\tau$. Similarly to Haasler et al. [12] introduce the following definitions:

$$\widehat{\phi}_t := (A^{t-1})^T \operatorname{diag}(u_{t-1}) \dots (A^0)^T \operatorname{diag}(u_0)\rho_0,$$
$$\phi_t := A^t \operatorname{diag}(u_{t+1}) \dots \operatorname{diag}(u_{\tau-1}) A^{\tau-1} \operatorname{diag}(u_\tau), \tag{18}$$

for $t = 0, \dots, \tau$. By setting $\widehat{\phi}_0 = \rho_0$ and $\phi_\tau = \mathbb{1}$ we get the recursive expressions:

$$\phi_t = A^t(u_{t+1} \odot \phi_{t+1}), \quad t = 0, \dots, \tau-1,$$
$$\widehat{\phi}_t = (A^{t-1})^T(u_{t-1} \odot \widehat{\phi}_{t-1}), \quad t = 1, \dots, \tau. \tag{19}$$

Let us recall that $u_0 = \exp(\lambda_0)$ and $u_t = \exp(B^T\lambda_t)$ for $t \geq 1$. With this notation, the second term of (15) is $-(\phi_t \odot \widehat{\phi}_t)^T u_t$, thus taking the derivative of the objective function (15) with respect to λ_t and setting it to zero we obtain:

$$\rho_0 = \phi_0 \odot u_0 \odot \widehat{\phi}_0,$$
$$\rho_t = B\phi_t \odot Bu_t \odot B\widehat{\phi}_t, \quad t \geq 1. \tag{20}$$

We observe that $Bu_t = B\exp(B^T\lambda_t) = \exp(\lambda_t)$. Therefore from (20) we can extract an update rule for λ_t. As the objective function is continuously differentiable and the problem is convex, the block coordinate ascent method converges Bertsekas [3, Proposition 2.7.1]. □

Given the limit point of Algorithm 1, the estimates for the marginals of the mass transfer matrices can be obtained as

$$M^t\mathbb{1} = \phi_t \odot \widehat{\phi}_t \odot u_t, \quad t = 0, \dots, \tau-1,$$
$$(M^{\tau-1})^T\mathbb{1} = \phi_\tau \odot \widehat{\phi}_\tau \odot u_\tau. \tag{21}$$

This can be seen by using the expression for the mass transfer matrices (10)–(12) together with the recursions (16), (17).

4. Modelling water flow as a Markov chain

We want to model the transition of pollution inside a water network as regulated by a Markov chain, in which the states are pipe segments. Pollution is assumed to be homogeneously diluted in the water of each segment, therefore we need to determine where the water of every pipe flows during each time interval. We start by considering a *line* network (*i.e.* without bifurcation), then we switch to the case of a generic water network.

4.1. Network without bifurcation

Let us consider a water network composed of N pipes that is indexed with $\mathcal{I} = \{1, \dots, N\}$. Without loss of generality, we assume that water flows according to the order of the pipes. We consider a discrete-time problem, with intervals of uniform length Δt. We assume that water enters the system from the first node, which can be considered a reservoir, and goes out from the last node, which has a positive demand. For the moment, we also assume that pipes do not split the network in multiple branches, *i.e.* water always flows from pipe i to pipe $i+1$, and they are all consecutive from the reservoir to the demand node.

We address the problem of finding the probability transition matrix of water moving through pipes. In general the system will not be stationary, since we allow demand to vary with time. Therefore, for each time step t we look for a transition matrix $A^t = [a^t_{ik}]_{i,k\in\mathcal{I}}$, in which a^t_{ik} is the probability for a random particle of water in pipe i at time t to go to pipe k at time $t+1$. Each pipe i has a fixed volume V_i. We also know the flow of water $F_i(t)$ (*i.e.* the volume of water passing through per time unit) in each pipe for all times t. We can transform it into a volume by multiplying it with the time unit. In this way we define the following quantity:

$$S_i(t) := \frac{F_i(t)\Delta t}{V_i}. \tag{22}$$

This quantity can be interpreted as the *speed of water* expressed in *pipe units*: if a pipe has $S_i(t) = 1$ it means that in the time interval $(t, t+\Delta t)$, the water that at time t is at the beginning of the pipe will reach exactly the end of the pipe. Using the speeds $S_i(t)$ we can compute the transition matrix A^t as specified in the following proposition.

Proposition 3. *Consider the sequence of pipes* $\mathcal{I} = \{1, \dots, n\}$ *with corresponding speed of water* $\{S_1, \dots, S_n\}$ *(in pipe units), and assume* *that* $\sum_{k=2}^{n} S_k^{-1} > 1$. *The proportion of water from pipe 1 that moves to pipe k is given by*

$$a_{1k} = \begin{cases} (1-\alpha)S_1/S_k & \text{if } k = n_1 < n_2 \\ S_1/S_k & \text{if } n_1 < k < n_2 \\ \beta S_1/S_k & \text{if } k = n_2, n_1 < n_2 \\ 1 & \text{if } k = n_1 = n_2 \\ 0 & \text{otherwise} \end{cases} \tag{23}$$

where the parameters $n_1, n_2 \in \mathbb{N}$ *and* $\alpha, \beta \in [0, 1)$ *are uniquely specified by the equations*

$$\sum_{k=1}^{n_1-1} \frac{1}{S_k} + \frac{\alpha}{S_{n_1}} = 1, \tag{24a}$$

$$\sum_{k=2}^{n_2-1} \frac{1}{S_k} + \frac{\beta}{S_{n_2}} = 1. \tag{24b}$$

□

Note that if $n_1 < n_2$ the water goes to more than one pipe and we need to use the first three cases in (23), while if $n_1 = n_2$ all the water goes to one pipe as specified by the forth case. Also, if $S_n < 1$, the water from pipe 1 cannot reach beyond pipe n in one time step, thus may be used as an absorbing state. Next, the Eq. (24a) can be solved simply by setting n_1 such that

$$\sum_{k=1}^{n_1-1} S_k^{-1} \le 1 < \sum_{k=1}^{n_1} S_k^{-1}, \text{ thus } \alpha = S_{n_1}\left(1 - \sum_{k=1}^{n_1-1} S_k^{-1}\right),$$

and similarly for (24b). Here we use the notation $\sum_{k=n}^{n-1} x_k = 0$, which means that the sum in (24a) is empty if $n_1 = 1$. Finally, note that by representing the flow in terms of speeds s_k, this proposition holds also if there is inflow of water into the system. Inflow changes the flow in the pipes, but this is represented by changes in the water speed, and as seen in the proof below, (23) still holds.

Proof. First note that the parameters n_1, n_2, α, β in (24) are uniquely determined since $\sum_{k=2}^{N} S_k^{-1} > 1$. This means that the water at the beginning of pipe 1 has reached a location in pipe n_1 corresponding to proportion α, and the water from the end of pipe 1 has reached a location in pipe n_2 corresponding to proportion β, and thus all water is distributed between those two points. Also note that the amount of water from pipe 1 that is in pipe k with $n_1 < k < n_2$ is inverse proportional to the speed S_k. This can be seen by noting that the proportion of water in pipe k that originated from pipe 1 is F_k/F_1, i.e., the quotient between the flows represent how much the water has been diluted by other incoming flows. Therefore, the amount of water in pipe k that originates from pipe 1 is $V_k F_k / F_1$ and by using (22) we get that $a_{1k} = S_1/S_k$. For pipes n_1 and n_2 the same reasoning holds except that the water only enters a proportion of the pipe corresponding to $(1-\alpha)$ and β, respectively. Finally, if $k = n_1 = n_2$ then all the water from pipe 1 goes to pipe k, thus $a_{1k} = (\beta - \alpha)S_1/S_k = 1$. This can also be seen from the Eq. (24), and by subtracting (24a) from (24b) and multiplying with S_1, we obtain

$$\frac{(1-\alpha)S_1}{S_{n_1}} + \sum_{k=n_1+1}^{n_2-1} \frac{S_1}{S_k} + \frac{\beta S_1}{S_{n_2}} = 1. \tag{25}$$

This represents the distribution of the water from pipe 1 into the pipes $n_1, n_1+1, \ldots, n_2$, and corresponds to the transitions in Eq. (23). □

In order to explain the procedure used in Proposition 3 we can look at the example in Fig. 1. Here we consider a simple line network composed of 5 pipes, whose speeds of water in pipe units S_i correspond respectively to $\frac{4}{3}, 3, 4, 4, 3$. We study the transition

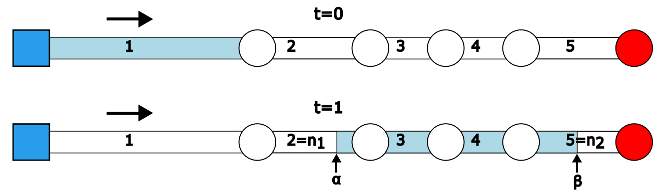

Fig. 1. A water network without bifurcation at two consecutive time steps. Water is flowing from a reservoir (blue square), to the demand node (red circle). In light blue we highlight the water in pipe 1 that transition according to the flow. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

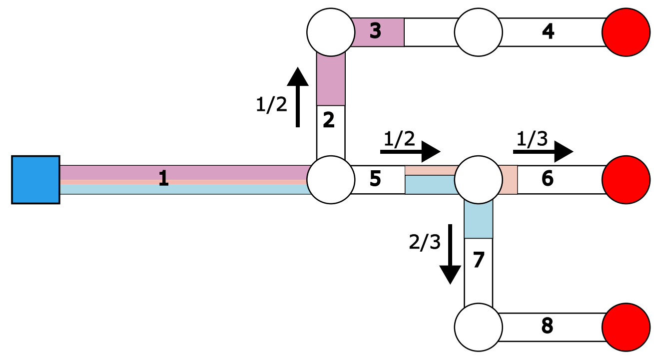

Fig. 2. A water network with bifurcations. Water is flowing from a reservoir (blue square), to the demand nodes (red circles). Water in pipe 1, after a time step flows to the rest of the network according to the coloured segments. At each splitting, the proportion of water that moves to each branch is indicated by the arrows. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

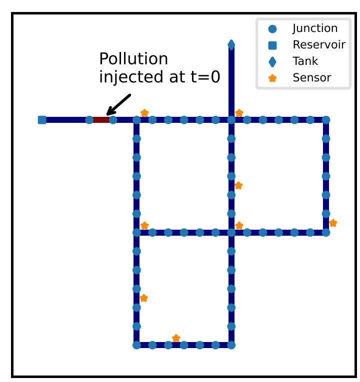

Fig. 3. Network and sensors.

of water in pipe 1, which is highlighted in light blue, in two consecutive time steps. We observe how the water from the first pipe fills pipes 3,4 and part of 2 and 5. Therefore we have that $n_1 = 2$, $n_2 = 5$, while $\alpha = \frac{3}{4}$ and $\beta = \frac{1}{2}$ indicate the fraction of the corresponding pipes where the water from pipe 1 starts and ends.

4.2. General case

In the case of splittings in the network, we consider all the paths a water particle can take in the network. This gives rise to a set of subproblems of the form described in Section 4.1, each of which is solved as described therein. Moreover, each path, i.e., each subproblem, is associated with a probability of a random water particle to take this path. The transition probability matrices A^t for each transition in the network are then found as a weighted sum of the solutions to the subproblems with the paths probabilities as weights.

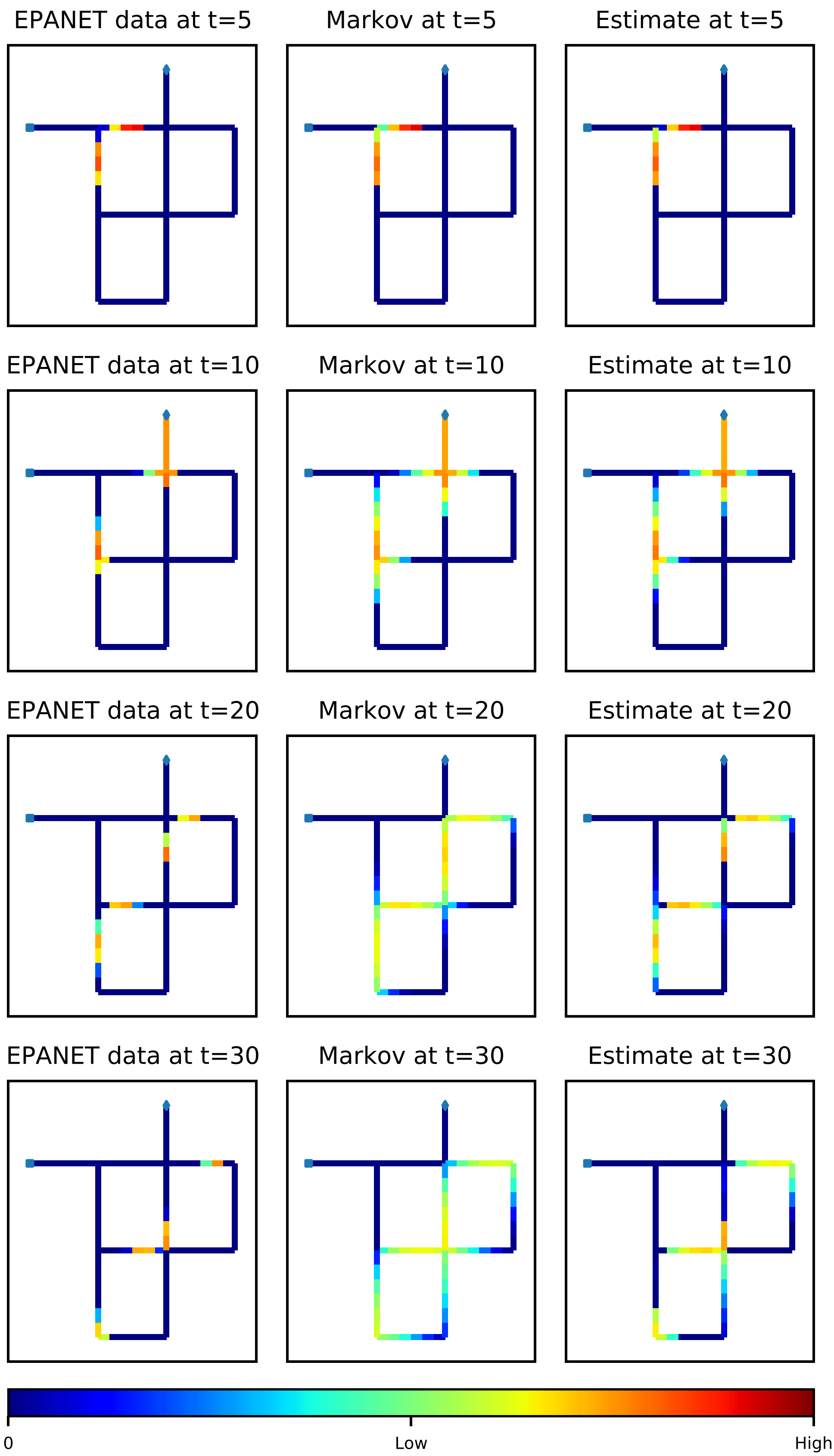

Fig. 4. Spread of pollution over a water network according to EPANET generated data (left), Markov propagation (center), Schrödinger bridge problem with sensor observation (right).

An example of this procedure is given in Fig. 2. Here we consider the transition of water in pipe 1 to the rest of the network at two consecutive time steps. This problem can be divided in three subproblems of the form introduced in Section 4.1. Each subproblem is indicated with a different colour. In violet we indicate the path formed by pipes {1,2,3,4}, which accounts for $\frac{1}{2}$ of the water. In pink we show the path {1,5,6}, whose weight is $\frac{1}{6}$. In light blue we show the subproblem {1,5,7,8}, that has weight $\frac{1}{3}$. The transition probabilities a_{1k} are then obtained as a weighted sum of the transition probabilities of each path.

5. Simulations

In this example we consider the problem of tracking the spread of pollution over a water network given the measurements from a set of sensors. In order to generate data, we use the water distribution system modelling software EPANET [28], developed by the US Environmental Protection Agency's Water Supply and Water Resources Division. We introduce an adapted version of the EPANET *net1* example, illustrated in Fig. 3. This model contains 65 states, corresponding to the pipes, and 8 sensors, positioned as in the figure. The flow is time varying, and water can leave the network in every node, according to a demand pattern. The transition probabilities a^t_{ik} are obtained using the method introduced in Section 4, based on the water flow from EPANET. We simulate the injection of pollution in a single pipe, as indicated in Fig. 3, during the first time step. We use the water quality analysis feature of EPANET to generate our reference data; decay or growth of the pollutant due to bulk or wall reactions are not considered. In Fig. 4 we propose a comparison firstly with a pure Markov model, *i.e.* assuming that the pollution propagates as $\mu_{t+1} = A^t\mu_t$; then we compare it with the estimate obtained by implementing the algorithm proposed in Proposition 2, exploiting the sensor observations. We run the algorithm until the violation of the primal constraints (4b)–(4e) is smaller that a tolerance (relative error with respect to the total amount of pollution in the system is smaller than 10^{-5}). From Fig. 4 we can observe how the Markov propagation results in a wider spread of pollution in the network, compared to the data generated by EPANET, in which pollution is more concentrated in a limited number of pipes. This behaviour is partly recovered in the estimate obtained with the methodology presented in Section 3: we can observe how the sensors measurements reduce the number of pipes containing pollution, obtaining a spread closer to the data.

6. Conclusions and further directions

In this work we have introduced a hidden Markov model framework for estimating the spread of pollution in a water network based on sparse sensor measurements. The proposed optimization problem is an extension of the Schrödinger bridge problem to the setting with partial observations. Building on this connection we derive an efficient algorithm for solving the problem.

In future work we aim to extend the method to also estimate the sources and amount of pollution, by relaxing the assumption that the initial distribution of pollution is known [8,17]. Bacterial contamination could be modelled more realistically by including dynamics for the bacteria growth into the model [4]. Moreover, we plan to investigate the problem of sensor placement [14]. Another open question is to quantify the robustness of the proposed framework to errors in the assumed Markov model.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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